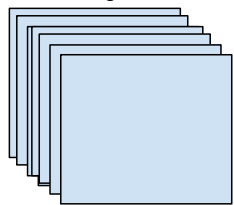


Prepare Terrain Segmentation Model

~31,707 Earth-based images



~134 Hand-Drawn labels

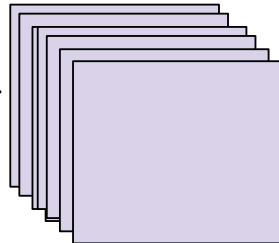


Learn features

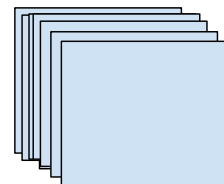
Pretrained DINOv2 ViT-S/14
(Classifier removed)

Custom Classifier
(Linear layer with
(indim=384, outdim=4))

~31,707 Pseudo Labels



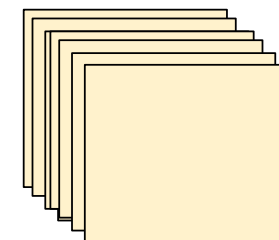
~31,707 Earth-based images



Fine tune our semantic Segmentation with learned masks

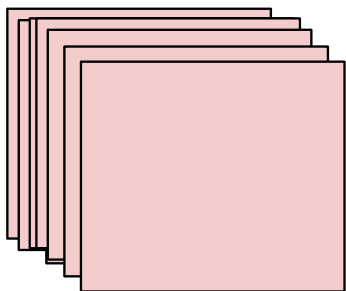
Pretrained DeepLabV3
(with ResNet-50 backbone)

Resulting Terrain Segmentation Masks



“Deploy” Model on Mars

Martian Test Images



Test using fine-tuned DeeplabV3

Pretrained DeepLabV3
(with ResNet-50 backbone)

Resulting Terrain Segmentation Masks

